Trainings

General Information

Note : For Dual Impeller water pumps and Jacket water pumps, use the following procedure. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html)

Engines with electronically actuated injectors do **not** have a separate low-temperature aftercooler water pump. Instead, the low-temperature aftercooler water pump element is located forward of the jacket water pump. Use the following procedure for more information about this pump. Refer to Procedure 008-062 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-062-tr.html)

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the temperature is below 50° C [120° F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

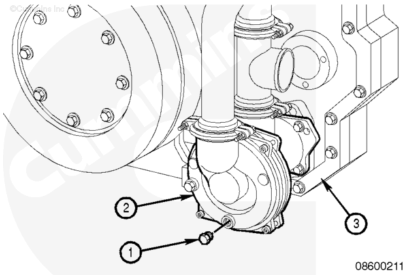
- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the air supply line to the starter to prevent accidental engine starting. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/56/56-012-022.html)

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

Remove

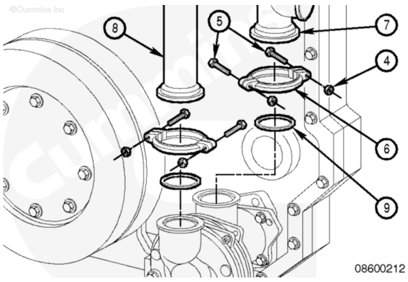
Note : The procedure for the QSK45 engine is shown. The procedure for the QSK60 engine is similar.

Remove the plug (1) on the low-temperature aftercooler water pump (2) to drain the remaining or trapped coolant from the engine (3).



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

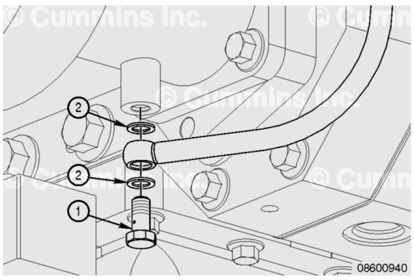
Do not support the weight of the water pump on the shaft end. Damage to the coupling or shaft can occur.

Remove the four nuts (4), capscrews (5), and the two clamps (6) from the inlet pipe (7) and bypass pipe (8).

Position the inlet pipe (7) and bypass pipe (8) aside to remove the three gaskets (9). Discard the old gaskets.

Remove the banjo capscrew (1) from the low-temperature aftercooler water pump.

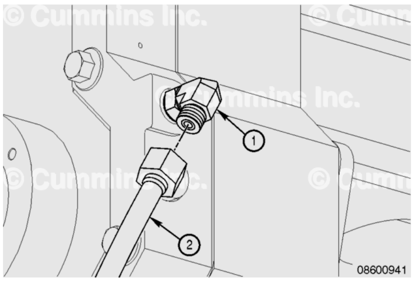
Discard the sealing washers (2).



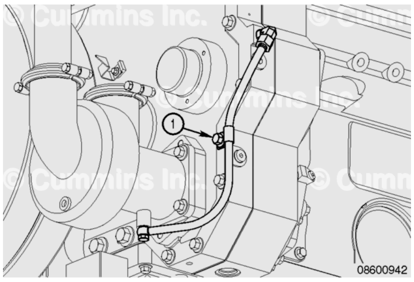
Note : Inspect the oil feed hose for damage. If replacement is required, remove as instructed in the

following steps.

Disconnect the oil feed hose (2) from the elbow (1) on the front gear housing.



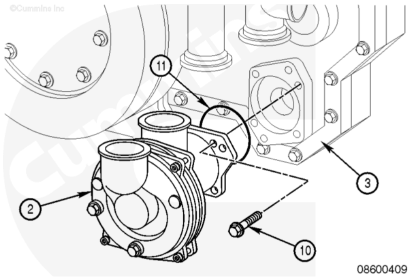
Remove the capscrew that attaches the oil feed hose P-clip (1) to the front gear housing and remove the oil feed hose.



Remove the four capscrews (10), the low-temperature aftercooler pump (2), and the o-ring (11) from the engine (3).

Inspect the o-ring for damage.

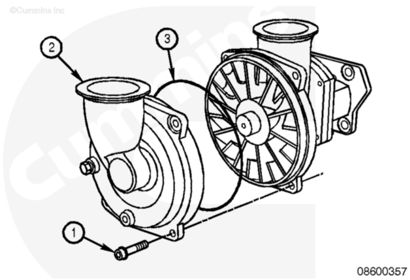
Discard the o-ring if damaged.



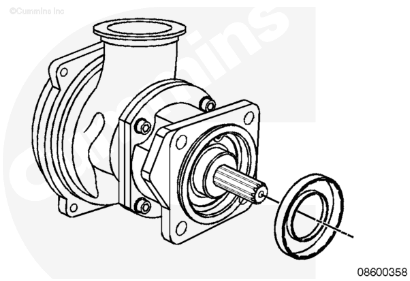
Disassemble

Remove the four capscrews (1) and separate the pump end body (2) from the drive end body.

Remove and discard the o-ring (3).

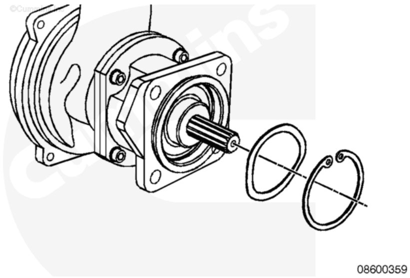


Turn the pump over to expose the splined shaft and remove the lip seal from the bearing housing.

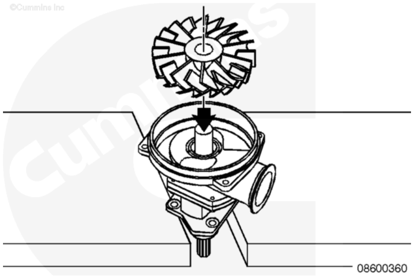


Remove the beveled retaining ring using snap ring pliers.

Remove the spring washer located under the snap ring.



There are two types of impellers. One impeller has three equally spaced threaded holes in the hub. This is the old style impeller. The new style has no holes in the impeller hub and the use of a hydraulic press is required to remove the impeller. The old style impeller can also be removed using a hydraulic press.



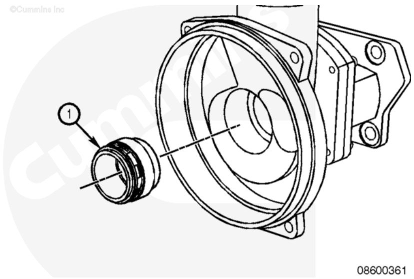
Support the bearing housing and press the shaft out of the impeller.

While removing the impeller, the drive end bearing will be pressed out of the bearing housing.

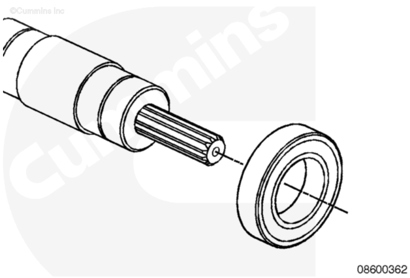
Once the impeller is free of the shaft, remove it from the press.

Continue to press the shaft to remove the drive end bearing from bearing housing.

Remove the rotating half of the mechanical seal (1) from the impeller end of the drive end body.

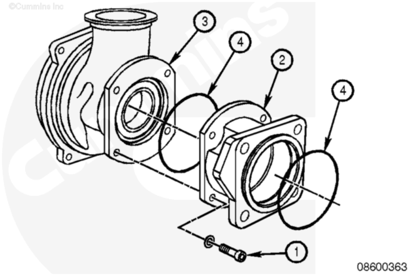


Press the drive end bearing off the shaft.

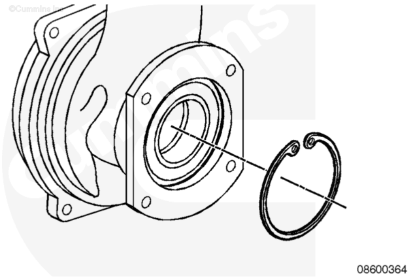


Remove the four capscrews (1) and separate the bearing housing (2) from the drive end body (3).

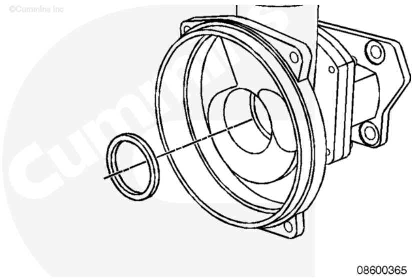
Discard the o-rings (4).



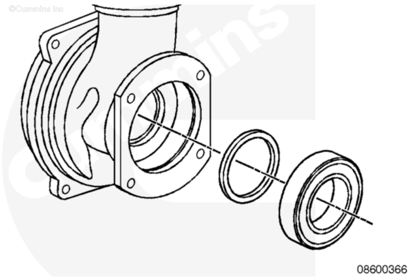
Remove the beveled retaining ring using snap ring pliers.



Remove the stationary half of the mechanical seal from the drive end body.



Press out the lip seal and bearing from the drive end body.

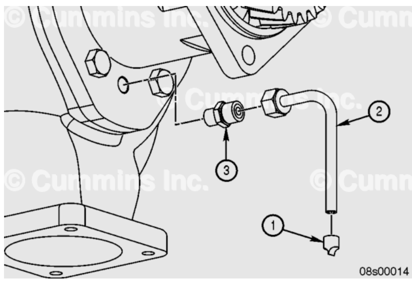


Duckbill:

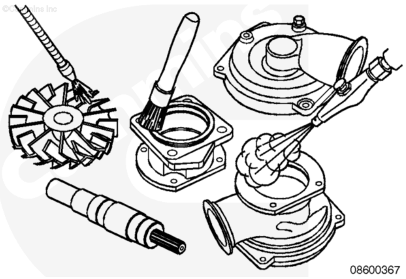
Note : There are no replacement parts or disassembly processes for the LTA water pump assembly.

Remove the filter (1) from the water pump weep hole drain tube.

Remove the drain tube (2) from the weep hole fitting (3).



Clean and Inspect for Reuse



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

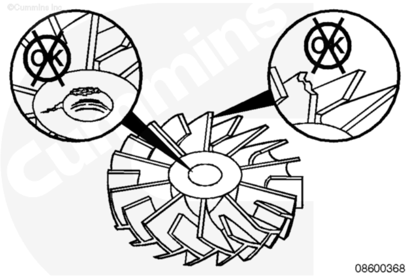
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

The mechanical seals, lip seals, o-rings, spring washers (lock washers), and bearings are **not** serviceable and **must** be replaced.

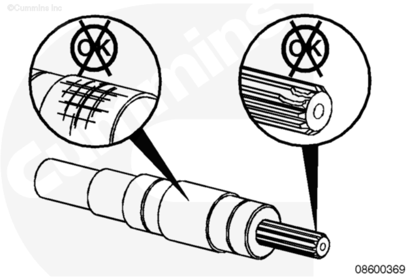
Clean all disassembled parts with solvent. Dry with compressed air.

The impeller **must** be replaced if the performance has deteriorated below an acceptable level. If the shaft has **not** suffered mechanical damage, it can be used again.

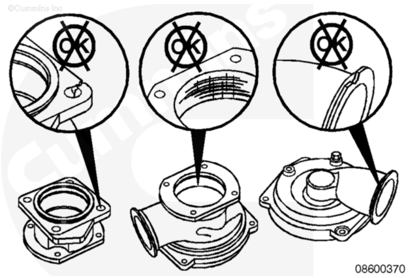
Inspect the impeller for damage or erosion. Replace if necessary.



Inspect the shaft for abnormal wear, pitting, or other damage. Replace if necessary.



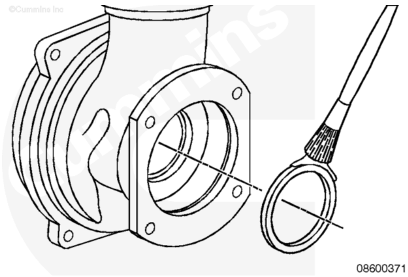
Inspect the pump housing and drive end housing for cracks or other damage. Replace if necessary.



Assemble

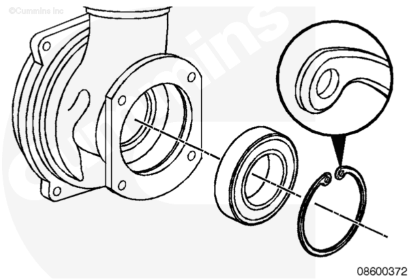
Install the lip seal into the drive end housing with the cup side of the seal toward the impeller.

Apply grease to the lip and upper face of the lip seal.



Press the bearing into the drive end housing.

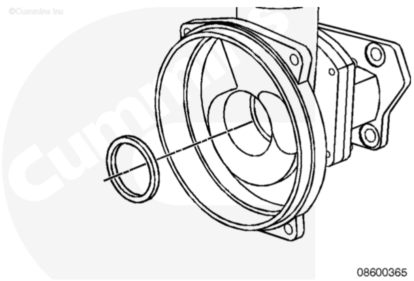
Install the beveled retaining ring with the bevel facing away from the bearing.



To operate correctly and effectively, the seal faces **must** be clean and dry.

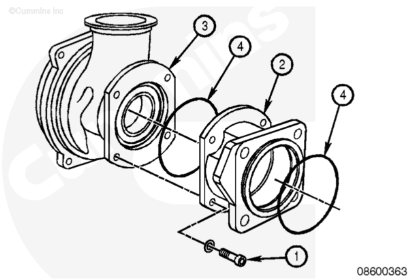
Apply a soapy solution to the stationary half of the mechanical seal and install the seal into the drive end housing from the impeller side.

Clean and dry the face of the seal with a lint-free cloth.

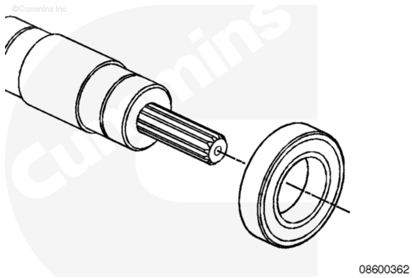


Install the bearing housing (2), with new o-rings (4), to the drive end body (3), and tighten the capscrews (1).

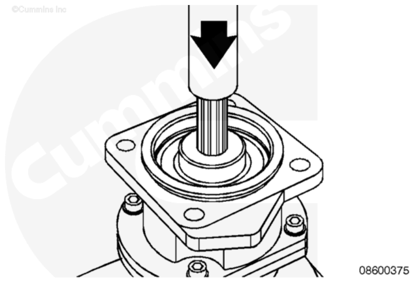
Torque Value: 33 n•m [24 ft-lb]



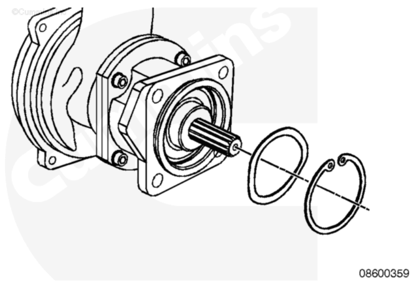
Press a new bearing onto the splined end of the shaft.



Install the shaft from the bearing housing end and press the shaft into the assembly until the bearing on the splined end bottoms out in the bearing housing.

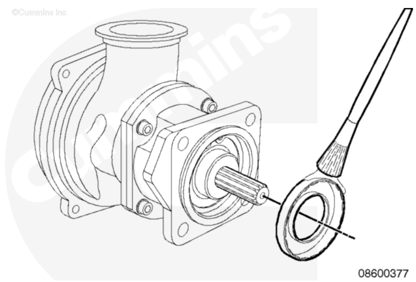


Install the spring washer and beveled retaining ring, with the bevel up, into the bearing housing.



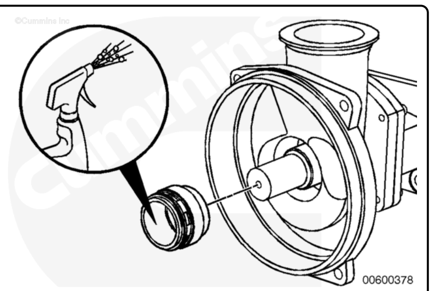
Apply grease to the bearing face and to the lip seal.

Install the lip seal into the bearing housing with the cup side of the seal toward the splined shaft.

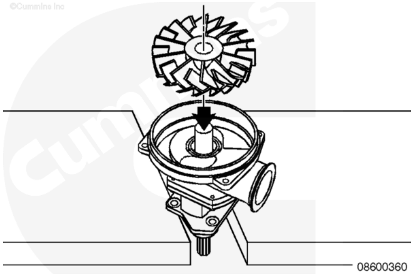


To operate correctly and effectively, the seal faces **must** be clean and dry.

Use a soapy solution on the inside of the rotating half of the mechanical seal, and press the seal over the shaft until it seats against the stationary half of the seal in the housing.



Support the splined shaft end of the pump and bearing housing and press the impeller onto the shaft.

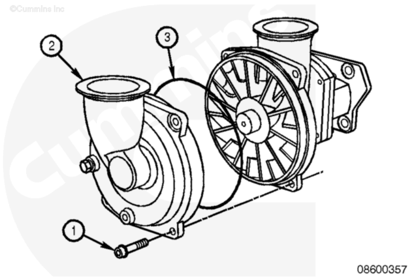


Install a new o-ring (3) on the drive end housing.

Install the end housing (2) and the capscrews and spring washers (1).

Tighten the capscrews in a crisscross pattern.

Torque Value: 33 n•m [24 ft-lb]



Duckbill:

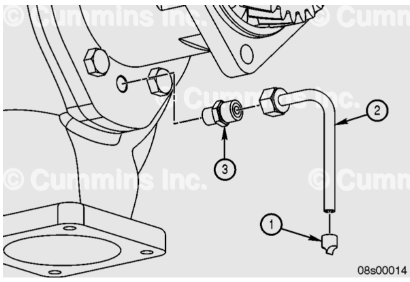
Install the weep hole drain tube fitting (3) onto the water pump and tighten.

Torque Value: 12 n•m [106+ in-lb]

Install the weep hole drain tube (2) onto the fitting (3) and tighten the compression nut.

Torque Value: 12 n•m [106+ in-lb]

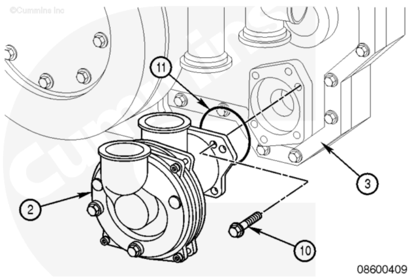
Install the filter (1) onto the weep hole drain tube (2).



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Do **not** support the weight of the pump on the shaft end.

Install the low-temperature aftercooler water pump (2), and o-ring (11), onto the engine (3) with the four capscrews (10).

Tighten the capscrews.

Torque Value: 80 n•m [59 ft-lb]

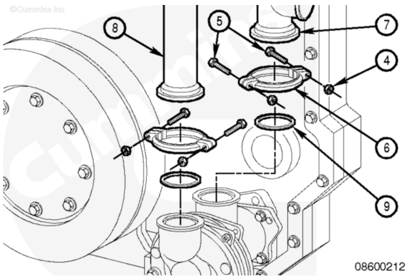
Position the two gaskets (9) on the low-temperature aftercooler water pump and bypass pipe (8) and inlet pipe (7).

Install the two clamps (6) on the bypass (8) and inlet (7) pipes with the four capscrews (5) and nuts (4).

Tighten the nuts.

Torque Value: 20 n•m [177 in-lb]

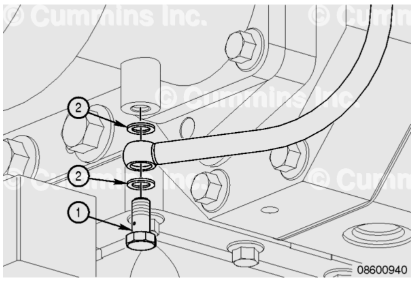
Note : Do not cause undue loading on the pump with the piping.



Install the banjo capscrew (1) and new sealing washers (2) to the low-temperature aftercooler water pump.

Tighten the banjo capscrew.

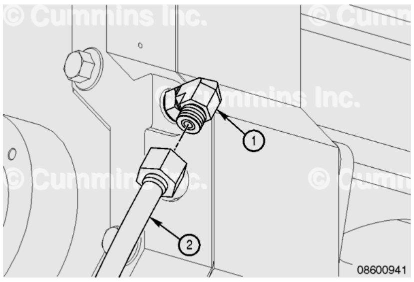
Torque Value: 37 n•m [27 ft-lb]



Connect the oil feed hose (2) to the elbow (1) on the front gear housing.

Tighten the hose fitting.

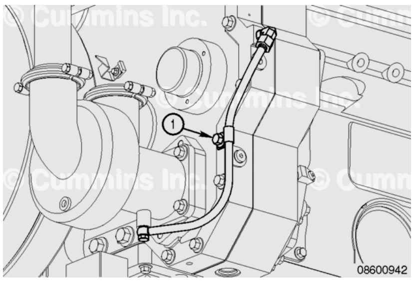
Torque Value: 16 n•m [142 in-lb]



Install the capscrew that attaches the oil feed hose and P-clip (1) to the front gear housing.

Tighten the capscrew.

Torque Value: 80 n•m [59 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Connect the air line to the starter. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/56/56-012-022.html)
- Operate the engine to 70°C [158°F] coolant temperature. Check for coolant leaks.

Last Modified: 27-Aug-2019